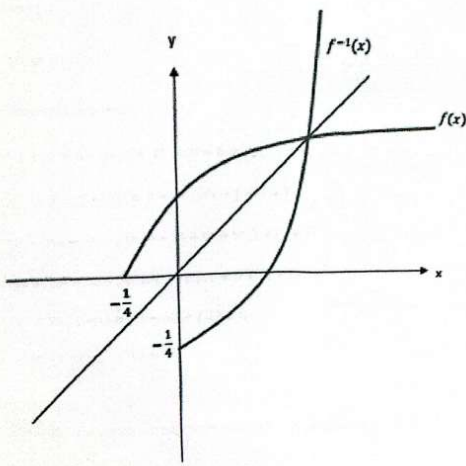
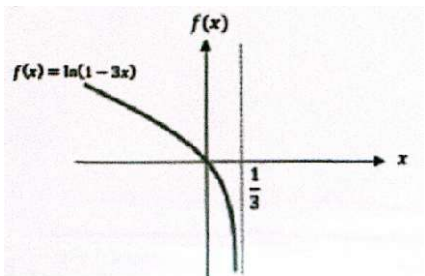


SM015 2014/2015 ANSWER SCHEME

NO	ANSWER SCHEME	MARKS
1	$3^x + 3^{(3-x)} = 12$ $3^x + 3^3 \cdot 3^{-x} = 12$ $3^x + \frac{3^3}{3^x} = 12$ $3^x + \frac{27}{3^x} = 12$ $\text{let } y = 3^x$ $y + \frac{27}{y} = 12$ $\frac{y^2 + 27}{y} = 12$ $y^2 - 12y + 27 = 0$ $(y - 9)(y - 3) = 0$ $y = 9, 3$ $3^x = 9 \quad 3^x = 3$ $x = 2 \quad x = 1$	K1 <

	$\left(1, \frac{7}{2}\right) \cup (6, \infty)$	J1
	TOTAL	6 MARKS
3(a) i)	$(f - g)(x) = f(x) - g(x)$ $= (2x + 1) - (x^2 + 2x - 1)$ $= -x^2 + 2$	K1 J1
ii)	$(3g - 2f)(x) = 3g(x) - 2f(x)$ $= 3(x^2 + 2x - 1) - 2(2x + 1)$ $= 3x^2 + 2x - 5$ $(3g - 2f)(1) = 3(1)^2 + 2(1) - 5$ $= 0$	K1 K1 K1 J1
3(b)	$f(x) = \sqrt{2x + \frac{1}{2}}$ $2x + \frac{1}{2} \geq 0$ $x \geq -\frac{1}{4}$ $D_f = \left[-\frac{1}{4}, \infty\right)$ $R_f = [0, \infty)$ 	K1 J1 J1 R1 R1 R1
	TOTAL	12 MARKS

4(a)	$\frac{\bar{z}}{ z ^2} = \frac{a - bi}{ a + bi ^2}$ $= \frac{a - bi}{\sqrt{(a^2 + b^2)}^2}$ $= \frac{a - bi}{a^2 + b^2}$ $= \frac{a - bi}{a^2 + b^2} \left(\frac{a + bi}{a + bi} \right)$ $= \frac{a^2 + abi - abi - (bi)^2}{(a^2 + b^2)(a + bi)}$ $= \frac{a^2 - b^2 i^2}{(a^2 + b^2)(a + bi)}$ $= \frac{a^2 + b^2}{(a^2 + b^2)(a + bi)}$ $= \frac{1}{a + bi}$ $= \frac{1}{z}$	K1 K1 K1 J1
4(b)	$\bar{z} = -z$ $a - bi = -(a + bi)$ $a - bi = -a - bi$ $a = -a$ $2a = 0$ $a = 0$ since $z = a + bi$ $= 0 + bi$ $= bi$ x is a complex number with only an imaginary part	K1 K1 J1
4(c)	$x(2 - i) = (\bar{z} + 1)(1 + i)$ $(a + bi)(2 - i) = [(a - bi) + 1](1 + i)$ $2a + 2bi - ai - bi^2 = (a - bi)(1 + i) + (1 + i)$ $2a + b + (2b - a)i = a - bi + ai + b + 1 + i$ $2a + (2b - a)i = (a + 1) + (a - b + 1)i$ by equating real part,	K1 K1

	The solution set : $\left\{x : x < \frac{5}{2} \cup x > 4\right\}$	J1
	TOTAL	13 MARKS
6(a)	$f[g(x)] = f\left[\frac{4x+1}{x}\right]$ $= \frac{1}{\left(\frac{4x+1}{x}\right) - 4}$ $= \frac{1}{\frac{4x+1-4x}{x}}$ $= x$ Since $f[g(x)] = x$ thus f and g are inverse function of each other	K1 K1 K1 J1 JI
6(b) i)	$f(x) = \ln(1-3x)$ $D_f : 1-3x > 0$ $1 > 3x$ $x < \frac{1}{3}$ $D_f = \left\{x : x < \frac{1}{3}\right\}$ $R_f = \{y \in \mathbb{R}\}$ 	K1 J1 J1 R1 R1 R1
6(b) ii)	$f[f^{-1}(x)] = x$ $\ln[1-3f^{-1}(x)] = x$ $1-3f^{-1}(x) = e^x$ $f^{-1}(x) = \frac{1-e^x}{3}$ $D_{f^{-1}} = \{x \in \mathbb{R}\}$ $R_{f^{-1}} = \left\{y : y < \frac{1}{3}\right\}$	K1 J1 J1 J1
	TOTAL	15 MARKS

NO	ANSWER SCHEME	MARKS
7	$y = x^3 - 4x^2 + 4x$ $\frac{dy}{dx} = 3x^2 - 8x + 4$ $3x^2 - 8x + 4 = 0$ $(3x - 2)(x - 2) = 0$ $x = \frac{2}{3}, x = 2$ $\frac{d^2y}{dx^2} = 6x - 8$ When $x = \frac{2}{3}, \frac{d^2y}{dx^2} = 6\left(\frac{2}{3}\right) - 8 = -4 (< 0)$ maximum When $x = 2, \frac{d^2y}{dx^2} = 6(2) - 8 = 4 (> 0)$ minimum $\therefore$ Relative maximum is $\left(\frac{2}{3}, \frac{32}{27}\right)$ $\therefore$ Relative minimum is $(2, 0)$	K1 Differentiate J1 Answer K1 Differentiate K1 Test J1 Answer J1 Answer
	TOTAL	6 MARKS
8	$\frac{dx}{dt} = -80, \frac{dy}{dt} = -100$ $z^2 = x^2 + y^2$ $z^2 = (0.15)^2 + (0.08)^2$ $z = 0.17$ $2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$ $z \frac{dz}{dt} = x \frac{dx}{dt} + y \frac{dy}{dt}$ $(0.17) \frac{dz}{dt} = (0.15)(-80) + (0.08)(-100)$ $\frac{dz}{dt} = -117.65$	B1 B1 Phytagoras K1 Find z K1 Differentiate J1 Answer K1 Substitute J1 Answer
	TOTAL	7 MARKS

NO	ANSWER SCHEME	MARKS
9(a)	$\lim_{x \rightarrow -3} \frac{x+3}{x^3+27}$ $\frac{x^2-3x+9}{x+3} \overline{) x^3+0x^2+0x+27}$ $\underline{-(x^3+3x^2)}$ $-3x^2+0x$ $\underline{-(-3x^2-9x)}$ $9x+27$ $\underline{-(9x+27)}$ 0 $\lim_{x \rightarrow -3} \frac{x+3}{(x+3)(x^2-3x+9)}$ $\lim_{x \rightarrow -3} \frac{1}{x^2-3x+9}$ $\frac{1}{(-3)^2-3(-3)+9}$ $= \frac{1}{27}$	K1 Long division K1 Simplify J1 Answer
9(b)	$\lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{x^2-9}}$ $\lim_{x \rightarrow -\infty} \frac{\frac{2x}{-x} - \frac{1}{-x}}{\sqrt{\frac{x^2}{x^2} - \frac{9}{x^2}}}$ $\lim_{x \rightarrow -\infty} \frac{-2 + \frac{1}{x}}{\sqrt{1 - \frac{9}{x^2}}}$ $= -2$	K1 Method K1 Simplify J1 Answer

NO	ANSWER SCHEME	MARKS
9(c)	$\lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{\sqrt{x} - 2}$ $\lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{\sqrt{x} - 2} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2}$ $\lim_{x \rightarrow 4} \frac{(x^2 - 3x - 4)(\sqrt{x} + 2)}{x - 4}$ $\lim_{x \rightarrow 4} \frac{(x - 4)(x + 1)(\sqrt{x} + 2)}{x - 4}$ $= (4 + 1)(\sqrt{4} + 2)$ $= 20$	K1 Conjugate K1 Factorise K1 Substitute J1 Answer
	TOTAL	10 MARKS
10(a)	$f(0) = 1 + e^0 = 2$ $\lim_{x \rightarrow 0^+} \frac{x + 6}{3 - x} = \frac{0 + 6}{3 - 0} = 2$ $\lim_{x \rightarrow 0^-} 1 + e^x = 1 + e^0 = 2$ $f(0) = \lim_{x \rightarrow 0} f(x) = 2$ $\therefore$ continuous at $x = 0$	K1 Substitute K1 Both J1 Answer K1 J1 Conclusion
10(b)	$f(4) = \lim_{x \rightarrow 4^+} C = \lim_{x \rightarrow 4^-} \frac{x + 6}{3 - x}$ $\frac{4 + 6}{3 - 4} = C = \frac{4 + 6}{3 - 4}$ $C = -10$ Continuous if $C = -10$ $\therefore$ discontinuous if $C \neq -10$	K1 3 conditions K1 Find C J1 Answer
10(c)	VA : $3 - x = 0$ $x = 3$ $\lim_{x \rightarrow 3^+} \frac{x + 6}{3 - x} = -\infty$ $\lim_{x \rightarrow 3^-} \frac{x + 6}{3 - x} = +\infty$ $\therefore$ Thus Vertical Asymptote is $x = 3$	K1 J1 Answer J1 Answer J1 Conclusion
	TOTAL	12 MARKS

NO	ANSWER SCHEME	MARKS
11(a)	$x = (\cos \theta)^3$ $\frac{dx}{d\theta} = 3(\cos \theta)^2 (-\sin \theta)$ $y = (\sin \theta)^3$ $\frac{dy}{d\theta} = 3(\sin \theta)^2 (\cos \theta)$ $\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$ $= 3(\sin \theta)^2 (\cos \theta) \cdot \frac{1}{3(\cos \theta)^2 (-\sin \theta)}$ $= -\frac{\sin \theta}{\cos \theta} = -\tan \theta$	K1 Differentiate K1 Differentiate K1 Chain Rule J1 Answer
11(b)	$(\cos \theta)^3 = \frac{\sqrt{2}}{4}$ $\cos \theta = 0.7071$ $\theta = \frac{1}{4}\pi, \frac{7}{4}\pi$ When $\theta = \frac{1}{4}\pi \Rightarrow \frac{dy}{dx} = -\tan\left(\frac{1}{4}\pi\right)$ $= -1$ When $\theta = \frac{7}{4}\pi \Rightarrow \frac{dy}{dx} = -\tan\left(\frac{7}{4}\pi\right)$ $= 1$	K1 J1 Answer K1 Substitute J1 Answer
11(c)	$\frac{d^2y}{dx^2} = -\sec^2 \theta \cdot \frac{1}{3(\cos \theta)^2 (-\sin \theta)}$ $= -\frac{1}{\cos^2 \theta} \cdot \frac{1}{3(\cos \theta)^2 (-\sin \theta)}$ $= -\frac{1}{3 \cos^4 \theta \sin \theta}$ $\frac{d^2y}{dx^2} = -\frac{1}{3 \cos^4 \left(\frac{\pi}{3}\right) \sin \left(\frac{\pi}{3}\right)} = \frac{32\sqrt{3}}{9}$	K1 Differentiate K1 J1 Answer K1 Substitute J1 Answer
	TOTAL	13 MARKS

NO	ANSWER SCHEME	MARKS
12(a)	$g'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{1-x-h} - \sqrt{1-x}}{h}$ $= \lim_{h \rightarrow 0} \frac{\sqrt{1-x-h} - \sqrt{1-x}}{h} \cdot \frac{\sqrt{1-x-h} + \sqrt{1-x}}{\sqrt{1-x-h} + \sqrt{1-x}}$ $= \lim_{h \rightarrow 0} \frac{1-x-h - (1-x)}{h(\sqrt{1-x-h} + \sqrt{1-x})}$ $= \lim_{h \rightarrow 0} \frac{-h}{h(\sqrt{1-x-h} + \sqrt{1-x})}$ $= \lim_{h \rightarrow 0} \frac{-1}{\sqrt{1-x-h} + \sqrt{1-x}}$ $= -\frac{1}{2\sqrt{1-x}}$	B1 Formula K1 Conjugate K1 Simplify K1 Substitute J1 Answer
12(b)	$e^y + xy + \ln(1+2x) = 1$ $e^y \frac{dy}{dx} + y(1) + x \frac{dy}{dx} + \frac{2}{1+2x} = 0$ $e^y \frac{dy}{dx} + y + x \frac{dy}{dx} + \frac{2}{1+2x} = 0$ $e^y \frac{dy}{dx} + y + x \frac{dy}{dx} + 2(1+2x)^{-1} = 0$ $\left(\frac{dy}{dx}\right)\left(e^y \frac{dy}{dx}\right) + (e^y)\left(\frac{d^2y}{dx^2}\right) + \frac{dy}{dx} + \left(\frac{dy}{dx}\right)(1) + (x)\left(\frac{d^2y}{dx^2}\right)$ $-2(1+2x)^{-2}(2) = 0$ $e^y \left(\frac{dy}{dx}\right)^2 + e^y \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + x \frac{d^2y}{dx^2} - \frac{4}{(1+2x)^2} = 0$ $(e^y + x) \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx}\right)^2 + 2 \frac{dy}{dx} - \frac{4}{(1+2x)^2} = 0$ At (0,0) $\Rightarrow$ $e^0 \frac{dy}{dx} + 0 + (0) \frac{dy}{dx} + 2(1+2(0))^{-1} = 0$ $\frac{dy}{dx} = -2$	K1 Differentiate all terms J1 2 correct J1 All correct K1 Differentiate all terms K1 Differentiate all terms J1 2 correct J1 All correct K1 Substitute

	At $(0,0) \Rightarrow$ $(e^0 + 0)\frac{d^2y}{dx^2} + e^0(-2)^2 + 2(-2) - \frac{4}{(1+2(0))^2} = 0$ $\frac{d^2y}{dx^2} + 4 - 4 - 4 = 0$ $\frac{d^2y}{dx^2} = 4$	K1 Substitute J1 Answer
	TOTAL	15 MARKS